

Developing a VR Meditation Program for College Students to Use on Campus with a Massage Chair

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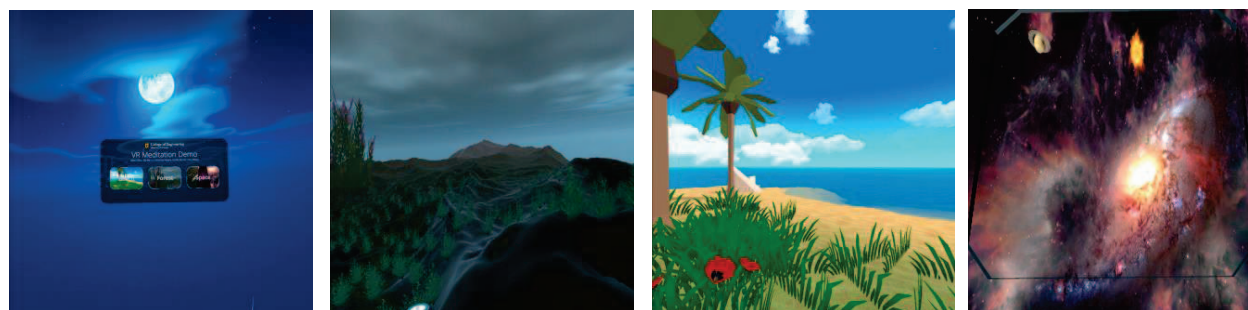


Fig. 1: VR Meditation application from left to right: (a) Main menu (b) Rain Forest (c) Low Poly Beach (d) Floating in Space

Abstract— College students often experience high level of stress due to academic commitment and associated factors. To explore the potential benefits of VR in stress management for students while they are on campus, we developed a VR meditation application to use with a massage chair. The application utilizes hand tracking technology and is designed to be used comfortably in conjunction with a massage chair to provide a convenient and accessible way to help manage stress for students while on campus. We plan to further develop the application and conduct further studies on the effects of this approach on students' mental health.

Keywords: Virtual Reality, Meditation, Education

I. INTRODUCTION

A large percentage (66%) of college students are reported to experience stress in a recent Gallup poll that surveyed more than 2,400 college students in March 2023 [1]. Moreover, studies reported college students feeling more anxious, depressed, fatigued, and distressed than prior to the COVID-19 pandemic. [2] University counseling centers are seeing an uptick in the number of students seeking help and are looking for innovative ways to address the students' stress. VR relaxation was shown to be effective in reducing subjectively reported stress in individuals with high stress [3] and can reduce psychological stress and is perceived by young adults as a useful and time-effective strategy to reduce psychological stress [4]. In addition, studies have shown that massage chairs could be used to effectively manage chronic stress [5]. Recently, more studies [6-8] have provided insight in the design of VR meditation. Our project is to develop a meditation program that takes advantage of both the VR technology and a massage chair to provide a convenient, accessible and cost-effective way for college students to alleviate the daily stresses while on campus.

II. METHODOLOGY

Our team utilized the Unity3D Engine to craft a captivating VR experience that blends serene environments with intuitive hand-

tracking interaction and deployed the application on Oculus Quest 2 headsets. Currently, there are three VR experiences developed as illustrated in Fig. 1, including beach, forest, and space. The Unity terrain builder played a pivotal role in shaping the diverse landscapes, from the verdant depths of a rainy forest to a sandy beach. Each terrain was crafted with a unique shader, ensuring that the rain forest and the water looked realistic.

Recognizing the relatively low levels of user interaction required for a relaxation VR application as well as the limitations of the traditional controllers in a massage chair setting, we eliminated the controller usage and fully relied on the hand-tracking technology provided by the headsets. By integrating Unity XR Interaction Toolkit, we seamlessly enabled hand-tracking within the application, empowering users to interact with their surroundings effortlessly. A UI menu was implemented through a virtual watch on wrist as illustrated in Fig. 2(a). Using their hands only, users can navigate through various scenes from the menu. This intuitive gesture-based control system proved to be a gamechanger, allowing users to fully immerse themselves in the experience without compromising their relaxation.



(a) Hand Tracking

(b) VR headset on a massage chair

Fig. 2: VR meditation set up.

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III. RESULTS

The VR meditation set up is illustrated in Fig. 2 (b). Eleven college students age between 17-30 experienced the VR meditation program in the massage chair located on campus for 5-20 minutes based on participants' preference. To gather valuable insights and feedback, a pre-survey and a post-survey were conducted among the participants before and after engaging with the program.

The pre-survey consisted of seven questions, addressing various aspects related to the participants demographics, prior experience with VR and meditation, stress levels, and goals for using the app. The survey results showed prior to using the program, 36% participants reported high to very high levels of stress, and 60% with moderate levels of stress. Most participants had little to no prior experience with VR meditation apps (82%), while the majority had experience with massage chairs (82%), indicating the potential novelty and synergy between the app and physical relaxation methods. Stress reduction was the primary goal for a significant portion of participants (60%) in using the program.

The post-survey inquired about the duration of usage, likelihood of future usage, physical sensations, clarity of VR scenes, side effects, and usefulness for meditation practice. 75% of the participants spent 5-10 minutes using the program, while 25% spent 10-20 minutes. This information provides a baseline for us to design the deployment of the program on campus, i.e. with a 20 min session for each student. The program was received favorably by the majority of the participants, with only one participant indicated extremely unlikely to use the program, and no negative side effects were reported.

As for the effectiveness of the program, most participants found the VR meditation experience relaxing (87.5%) and reported feeling more focused and present afterward (75%). All participants (100%) indicated that the VR meditation app helped reduce their stress levels (Table 1).

Table 1: Insights into User Satisfaction and Experiences after using VR Meditation App

Survey questions	Percentage
Found the experience relaxing	87.5%
Felt more focused and present	75%
The program helped with stress reduction	100%
Experience negative side effects	0%
Found the program to be a useful tool for meditation	62.5%

Additionally, participants were asked to provide feedback and suggestions for improvement.

"All scenes were well done and the hand tracking integration was the coolest in my experience"

"Rainy forest is really cool and I observed many small details like fire flies and particularly the music was good in the background."

"I like the hand tracking and menu button on the watch. I wasn't sure what the tan object to the right of the user in the space scene was."

"Would have been better on my own. Better without the chair. Massage chair is uncomfortable. I like all the scenes but would like to see them more developed."

"On the beach scene, I saw a little bit of tearing in the sky when the massage chair would cause my head to move quickly. It didn't cause any problems but was a little annoying."

"Probably reducing the brightness of the beach scene would enhance the experience."

The participants feedback indicate that the virtual scene design can be further improved; and not everyone enjoys the massage chair which can be made optional during the VR session. In addition, with the rapid adoption of VR headset among student population, the VR portion of the program can be made available to students for them to use at home.

IV. CONCLUSION

A VR meditation program has been designed to be used on campus with a massage chair by college students. Careful consideration has been given to the design of the program, including the scenes selection, hand tracking utilization, and UI menu implementation. The program was tested by 11 college students and most of the feedback is very favorable. It's promising for the team to further develop the program and study its impact on college students stress reduction both on campus or at home.

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